

STAGE FIVE • THINKING, REASONING & MENTAL MODELS

FIRSTS

FACILITATOR GUIDE

Learning objectives, talking points, and misconceptions to preempt for all 30 FIRSTS

A Citrus Edition Facilitator Resource • Companion to the FIRSTS Manual

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FIRSTS • Facilitator Guide

Stage Five • Thinking, Reasoning & Mental Models

HOW TO USE THIS GUIDE

This guide is for whoever is facilitating Stage Five, live or on video, individually or in a cohort. It sits alongside the FIRSTS Manual and Templates workbook — it does not replace them.

Stage Five spans two source categories — G: Thinking, Reasoning & Mental Models (foundational) and H: Thinking Frameworks (advanced) — thirty FIRSTS total. This guide organizes them into four sessions: Foundational Mental Models (G1–G8), Applied Reasoning & Reflection (G9–G15), Advanced Frameworks I (H16–H23), and Advanced Frameworks II (H24–H30).

This stage is genuinely dense and abstract compared to earlier stages — nearly every FIRST asks participants to apply a reasoning tool to a real, current problem rather than reflect on themselves directly. Push consistently for real, specific examples; abstract discussion of a mental model without a genuine application produces very little retention.

SESSION 1

Foundational Mental Models

Covers *FIRSTS G1–G8*

FIRST G1

Mental Model Awareness

LEARNING OBJECTIVE

Participants can name at least three mental models they already use, even if unconsciously, and identify one they'd like to use more deliberately.

KEY TALKING POINTS

- Reasoning inconsistently, sometimes carefully, sometimes on pure instinct, is the default without conscious awareness of the tools you're using.
- Naming a mental model is the first step toward using it deliberately rather than accidentally.
- This is the foundation for the entire stage — everything that follows assumes this baseline awareness.

COMMON MISCONCEPTIONS TO PREEMPT

- Treating mental models as trivia to memorize rather than tools to actually apply.
 - Trying to learn dozens of models at once instead of starting with a small, genuinely useful set.
-

SUGGESTED TIME: 20 min

FIRST G2

Cause-and-Effect Mapping

LEARNING OBJECTIVE

Participants map a real recent outcome back through its actual causal chain, not just the first cause that comes to mind.

KEY TALKING POINTS

- Without deliberately tracing cause and effect, it's easy to repeat mistakes or credit success to the wrong factor.
- The real mechanism often stays invisible unless you deliberately dig for it.
- Correlation and causation get confused constantly — flag this explicitly as a specific trap to watch for.

COMMON MISCONCEPTIONS TO PREEMPT

- Stopping at the first cause instead of tracing back further.
 - Assuming correlation is causation without checking the actual mechanism.
-

SUGGESTED TIME: 20 min

FIRST G3

Problem Breakdown Exercise

LEARNING OBJECTIVE

Participants take one genuinely overwhelming problem and break it into an ordered sequence of achievable steps.

KEY TALKING POINTS

- Complex problems feel paralyzing as one giant task; breaking them down converts overwhelm into sequence.
- This is often the actual difference between finishing and stalling out, not raw ability.
- Ordering matters as much as breaking down — a list tackled randomly loses much of its value.

COMMON MISCONCEPTIONS TO PREEMPT

- Breaking the problem down so finely the list itself becomes overwhelming.
- Skipping the ordering step and tackling parts randomly instead of sequentially.

SUGGESTED TIME: *20 min*

FIRST G4

Decision Matrix

LEARNING OBJECTIVE

Participants build a weighted decision matrix for one real current decision, comparing options against explicit criteria.

KEY TALKING POINTS

- Without structure, decisions often go to whichever option was considered most recently or most vividly, not the one that scores best.
- The matrix externalizes the comparison so it can be evaluated honestly, not just felt.
- Watch for weights that quietly favor a predetermined answer — that defeats the entire purpose.

COMMON MISCONCEPTIONS TO PREEMPT

- Choosing weights that quietly favor the option you already wanted.
- Including so many criteria the matrix becomes unwieldy and loses its clarifying power.

SUGGESTED TIME: *25 min*

FIRST G5

Root Cause Analysis

LEARNING OBJECTIVE

Participants trace one real recurring problem past its surface symptom to a genuine underlying cause.

KEY TALKING POINTS

- Fixing a symptom without addressing the root cause means the problem resurfaces in a slightly different form.

- The first answer is often a symptom, not a cause — push participants to keep asking why.
- This should be honest diagnosis, not self-blame — name that distinction explicitly.

COMMON MISCONCEPTIONS TO PREEMPT

- Stopping at the first answer, which is often a symptom rather than a genuine cause.
- Turning the exercise into self-blame instead of honest, dispassionate diagnosis.

SUGGESTED TIME: 20 min

FIRST G6

Critical Thinking Drill

LEARNING OBJECTIVE

Participants identify the unstated assumptions underneath one real claim they've recently encountered.

KEY TALKING POINTS

- Most claims, in coursework, media, or casual conversation, rest on unstated assumptions that go unexamined.
- Practicing this deliberately turns critical thinking into a reflex, not an occasional effort.
- The real test is applying equal scrutiny to claims you already agree with, not just ones you're inclined to challenge.

COMMON MISCONCEPTIONS TO PREEMPT

- Using critical thinking only to attack claims you already disagree with.
- Giving claims you personally like a pass without the same scrutiny.

SUGGESTED TIME: 20 min

FIRST G7

Hypothesis Testing

LEARNING OBJECTIVE

Participants design one small, genuine test for an assumption currently shaping a real decision.

KEY TALKING POINTS

- Untested assumptions quietly shape decisions constantly, and many turn out wrong when actually checked.
- A small experiment is a low-cost way to replace guessing with real evidence.
- The test needs to be genuinely capable of disproving the prediction, or it isn't a real test.

COMMON MISCONCEPTIONS TO PREEMPT

- Making the test so small or biased it can't actually disprove the prediction.
- Ignoring inconvenient results that contradict what you wanted to believe.

SUGGESTED TIME: 20 min

Scenario Planning

LEARNING OBJECTIVE

Participants map best-case, worst-case, and most-likely scenarios for one real upcoming decision.

KEY TALKING POINTS

- Planning only for the hoped-for outcome leaves you unprepared if things go differently.
- Planning only for the worst case can create unnecessary anxiety without genuine preparation benefit.
- The most-likely scenario is often the most useful, even though it feels less interesting than the extremes.

COMMON MISCONCEPTIONS TO PREEMPT

- Spending so much time on the worst case that it becomes catastrophizing rather than planning.
- Skipping the most-likely scenario because it feels less interesting than the extremes.

SUGGESTED TIME: 25 min

SESSION 2

Applied Reasoning & Reflection

Covers *FIRSTS G9-G15*

FIRST G9

Cognitive Bias Audit

LEARNING OBJECTIVE

Participants identify at least two of their own genuinely recurring cognitive biases from real past decisions.

KEY TALKING POINTS

- Everyone has biases, but most people assume their own reasoning is more objective than it actually is.
- Naming a specific, recurring bias turns an invisible pattern into something you can actually catch in the moment.
- This is meant to be an ongoing watch, not a one-time labeling exercise.

COMMON MISCONCEPTIONS TO PREEMPT

- Treating the audit as a one-time academic exercise rather than an ongoing thing to watch for.
 - Naming biases in the abstract without connecting them to real, specific past decisions.
-

SUGGESTED TIME: 20 min

FIRST G10

Systems Thinking Exercise

LEARNING OBJECTIVE

Participants map one real problem as a system with feedback loops, not just an isolated failure.

KEY TALKING POINTS

- Many problems that look like isolated failures are actually symptoms of how a larger system is structured.
- Fixing the visible symptom without seeing the system means the same issue tends to resurface elsewhere.
- A real system usually has more than one loop — push participants to keep looking past the first one found.

COMMON MISCONCEPTIONS TO PREEMPT

- Oversimplifying the map to just a few nodes when the real system has more interconnected parts.
 - Stopping at the first loop found without checking whether others exist.
-

SUGGESTED TIME: 25 min

FIRST G11

Feedback Integration

LEARNING OBJECTIVE

Participants identify one piece of real past feedback and name a specific, concrete change made because of it.

KEY TALKING POINTS

- Feedback that isn't integrated is functionally the same as no feedback at all.
- The value only appears once feedback actually changes what you do next, not when it's simply heard.
- Repeated feedback on the same point is a signal worth taking seriously, not dismissing as nagging.

COMMON MISCONCEPTIONS TO PREEMPT

- Hearing the same feedback repeatedly without noticing the pattern.
- Becoming defensive instead of curious about what's actually being pointed out.

SUGGESTED TIME: 20 min

FIRST G12

Strategic Prioritization

LEARNING OBJECTIVE

Participants rank a real current task list by genuine impact, not just ease or urgency.

KEY TALKING POINTS

- Without deliberate prioritization, low-impact but easy tasks often crowd out higher-impact but harder ones.
- Genuine prioritization requires real differentiation — not everything can be top priority.
- This connects directly to Root Cause Analysis and Systems Thinking earlier in this session; impact often traces back to root causes, not symptoms.

COMMON MISCONCEPTIONS TO PREEMPT

- Rating everything as high-impact because it all feels important in the moment.
- Confusing urgency with genuine importance when ranking tasks.

SUGGESTED TIME: 20 min

FIRST G13

Mental Model Application

LEARNING OBJECTIVE

Participants apply one specific mental model from earlier in this stage to a real current problem, in writing.

KEY TALKING POINTS

- Knowing about a mental model and actually applying it under real conditions are genuinely different skills.
- Writing out the reasoning, not just thinking it loosely, is what closes the gap between knowing and doing.
- This is a deliberate checkpoint tying together everything covered in Session 1.

COMMON MISCONCEPTIONS TO PREEMPT

- Applying the model loosely or informally without actually writing out the reasoning.
- Believing you've thought something through when informal application hasn't actually tested that.

SUGGESTED TIME: 25 min

FIRST G14

Pattern Recognition Drill

LEARNING OBJECTIVE

Participants identify one genuine pattern across three or more real past instances of a recurring issue.

KEY TALKING POINTS

- A single data point rarely tells you much; the same issue appearing three or four times reveals a genuine pattern.
- Patterns require paying close enough attention to notice them across scattered, separate instances.
- Overgeneralizing from too few data points is the opposite failure mode, equally worth watching for.

COMMON MISCONCEPTIONS TO PREEMPT

- Seeing a pattern in too few data points and overgeneralizing from coincidence.
- Dismissing a real pattern because no single instance seemed significant on its own.

SUGGESTED TIME: 20 min

FIRST G15

Reflection on Decision-Making

LEARNING OBJECTIVE

Participants review several of their own real past decisions and identify one recurring reasoning mistake.

KEY TALKING POINTS

- Without deliberate reflection, the same reasoning mistakes repeat across different decisions, each experienced in isolation.
- This closes Session 1's foundational models by turning individual practice into a genuine, visible pattern over time.
- An honest reflection includes decisions that went poorly, not just the ones worth celebrating.

COMMON MISCONCEPTIONS TO PREEMPT

- Journaling only about decisions that went well, producing a skewed, overly positive picture.
- Reviewing decisions in isolation rather than looking for a genuine pattern across several.

SUGGESTED TIME: *20 min*

SESSION 3

Advanced Frameworks I

Covers *FIRSTS H16-H23*

FIRST H16

SWOT Thinking Exercise

LEARNING OBJECTIVE

Participants build a genuine SWOT analysis for one real personal decision, not a generic business scenario.

KEY TALKING POINTS

- SWOT is usually taught as a business tool, but the same structure works well for personal decisions.
- It forces consideration of internal factors (strengths, weaknesses) alongside external ones (opportunities, threats), not just one side.
- Genuine specificity to the actual decision matters far more than a technically complete but generic list.

COMMON MISCONCEPTIONS TO PREEMPT

- Listing generic strengths and weaknesses without connecting them to the specific decision at hand.
 - Producing a list that feels thorough on paper but isn't genuinely useful for the actual decision.
-

SUGGESTED TIME: 20 min

FIRST H17

First-Principles Analysis

LEARNING OBJECTIVE

Participants strip one real assumption down to what's genuinely, verifiably required, rather than inherited from precedent.

KEY TALKING POINTS

- Most problem-solving happens by comparison to precedent, efficient but capable of trapping you in unnecessary assumptions.
- First-principles thinking strips those assumptions away to find what's truly, verifiably required.
- This is not the same as being contrarian — it's about genuine verification, not just disagreeing with convention.

COMMON MISCONCEPTIONS TO PREEMPT

- Confusing “first principles” with simply having a contrarian opinion.
 - Not actually verifying which assumptions are truly fundamental versus just unexamined habit.
-

SUGGESTED TIME: 25 min

FIRST H18

Inversion Thinking Drill

LEARNING OBJECTIVE

Participants list what would guarantee failure in a real current goal, then translate that list into preventive action.

KEY TALKING POINTS

- It's often easier to see clearly what would cause failure than to directly identify what guarantees success.
- Inversion uses that asymmetry deliberately to surface risks you might otherwise miss.
- The exercise isn't complete until the failure list is translated back into concrete, preventive action.

COMMON MISCONCEPTIONS TO PREEMPT

- Treating the exercise as pure pessimism without translating the failure list back into concrete action.
- Stopping at the list itself rather than using it to actually change a real plan.

SUGGESTED TIME: *20 min*

FIRST H19

Opportunity Cost Exercise

LEARNING OBJECTIVE

Participants name the genuine next-best alternative being traded away by a real current choice.

KEY TALKING POINTS

- Evaluating an option purely on its own merits, without naming what's traded away, makes almost anything look reasonable.
- Opportunity cost forces the comparison that actually matters: against the real next-best alternative.
- Comparing only against doing nothing understates the real cost significantly.

COMMON MISCONCEPTIONS TO PREEMPT

- Comparing the option only against doing nothing, rather than the actual next-best alternative.
- Ignoring genuine trade-offs because the chosen option feels obviously right in isolation.

SUGGESTED TIME: *20 min*

FIRST H20

80/20 Analysis

LEARNING OBJECTIVE

Participants analyze their own real time or effort data to find which small subset produces most of the actual results.

KEY TALKING POINTS

- Effort and results are rarely evenly distributed; without checking, it's easy to spread equal time across everything.
- A small subset of activities is often quietly doing most of the real work.
- The 80/20 split is a pattern to investigate honestly, not a literal, exact ratio to force onto the data.

COMMON MISCONCEPTIONS TO PREEMPT

- Applying the 80/20 split too literally as an exact ratio rather than a pattern to investigate.
- Assuming the same 20% applies universally without checking your own actual data.

SUGGESTED TIME: 20 min

FIRST H21

Systems Thinking Application

LEARNING OBJECTIVE

Participants apply systems thinking to a real team or project problem, tracing structural causes rather than individual blame.

KEY TALKING POINTS

- Personal-scale systems thinking (from G10) builds the skill; applying it to a real team is where it starts to genuinely change outcomes.
- Most team-level problems are systemic rather than individual, even when they initially look like one person's fault.
- The temptation to default back to blaming individuals grows stronger exactly when the mapping gets difficult — push through it.

COMMON MISCONCEPTIONS TO PREEMPT

- Defaulting back to blaming individuals once the systems mapping gets genuinely difficult.
- Treating a structural problem as a personal failing because that's the simpler explanation.

SUGGESTED TIME: 25 min

FIRST H22

Decision Tree Exercise

LEARNING OBJECTIVE

Participants build a visual decision tree for one real, complex, multi-step decision.

KEY TALKING POINTS

- Complex decisions with multiple sequential choices are hard to hold entirely in your head at once.
- A visual tree externalizes the structure so it can be evaluated clearly rather than losing track of branches.
- The tree should simplify the decision, not become as overwhelming as the original problem.

COMMON MISCONCEPTIONS TO PREEMPT

- Building a tree so detailed it becomes as overwhelming as the original decision.
 - Including every conceivable branch instead of focusing on the ones that genuinely matter.
-

SUGGESTED TIME: 25 min

FIRST H23

Scenario Analysis

LEARNING OBJECTIVE

Participants build a scaled-up scenario analysis for one genuinely complex decision, focused on the 2–3 variables that drive the most uncertainty.

KEY TALKING POINTS

- As decisions grow more complex, with more variables and stakeholders, simple scenario planning (from G8) needs to scale up.
- The goal is still a clear, actionable plan for each scenario, not just a longer list of possibilities.
- Focus on the two or three variables that actually drive the most uncertainty, not every conceivable factor.

COMMON MISCONCEPTIONS TO PREEMPT

- Adding so many variables the analysis becomes unmanageable.
 - Treating every possible factor as equally important rather than focusing on the ones that actually drive uncertainty.
-

SUGGESTED TIME: 25 min

SESSION 4

Advanced Frameworks II

Covers *FIRSTS H24-H30*

FIRST H24

Checklists & SOPs Creation

LEARNING OBJECTIVE

Participants create one real checklist or SOP for a task they perform repeatedly, concise enough to actually use.

KEY TALKING POINTS

- Repeated tasks done from memory alone accumulate small, inconsistent errors over time.
- A written checklist catches what memory alone tends to miss, especially under time pressure.
- Length is the enemy here — a checklist too long to actually use defeats its own purpose.

COMMON MISCONCEPTIONS TO PREEMPT

- Making the checklist so long and detailed it becomes a chore to actually use.
 - Treating comprehensiveness as more important than genuine usability.
-

SUGGESTED TIME: 20 min

FIRST H25

Root Cause Framework

LEARNING OBJECTIVE

Participants apply a named, structured root-cause method (such as Five Whys) to a real current problem.

KEY TALKING POINTS

- A named, structured method makes root cause analysis easier to apply consistently under pressure.
- This extends G5's Root Cause Analysis into a formal, repeatable framework rather than an ad hoc dig.
- Five whys is a guideline, not a rigid rule — stop when the true root cause is reached, whether that's at three whys or seven.

COMMON MISCONCEPTIONS TO PREEMPT

- Forcing exactly five whys even when the true root cause is reached sooner.
 - Stopping early at a comfortable answer before the root cause is genuinely reached.
-

SUGGESTED TIME: 20 min

FIRST H26

Feedback Loop Exercise

LEARNING OBJECTIVE

Participants design one deliberate try-observe-adjust loop for a real, ongoing skill or project.

KEY TALKING POINTS

- Most meaningful skills and projects improve through repeated cycles of trying, observing, and adjusting.
- Getting it right on the first attempt is the exception, not the expectation — build the loop deliberately instead.
- The loop only works if each cycle is actually recorded and reviewed, not just repeated.

COMMON MISCONCEPTIONS TO PREEMPT

- Repeating attempts without actually recording or reviewing what changed.
- Treating repetition alone as equivalent to genuine, deliberate iteration.

SUGGESTED TIME: 20 min

FIRST H27

Comparative Analysis

LEARNING OBJECTIVE

Participants compare 2-3 real options against the same explicit, shared set of criteria.

KEY TALKING POINTS

- Options often get compared unevenly, one on cost, another on convenience, another on gut feeling, which makes the comparison unreliable.
- Explicit, shared criteria applied to every option fix that unevenness.
- This directly extends the Decision Matrix from G4 into a sharper, criteria-first comparison.

COMMON MISCONCEPTIONS TO PREEMPT

- Using criteria so broad or vague that scoring becomes arbitrary rather than genuinely comparative.
- Comparing options on different, inconsistent dimensions rather than the same shared criteria.

SUGGESTED TIME: 20 min

FIRST H28

Mental Model Combination

LEARNING OBJECTIVE

Participants apply two mental models from this stage together to one real, genuinely complex problem.

KEY TALKING POINTS

- Real problems are rarely clean enough for one mental model to solve completely.
- Combining models often reveals insight that neither would surface alone.

- The combination should be genuinely called for by the specific problem, not performed just to seem sophisticated.

COMMON MISCONCEPTIONS TO PREEMPT

- Combining models just to seem sophisticated rather than because the problem genuinely calls for it.
- Forcing a combination onto a problem simple enough for a single model to handle well.

SUGGESTED TIME: 25 min

FIRST H29

Hypothesis-Driven Approach

LEARNING OBJECTIVE

Participants design a small-scale test for a core assumption before committing to a larger plan built on it.

KEY TALKING POINTS

- Committing fully to a large plan based on untested assumptions risks significant wasted effort if a core assumption is wrong.
- Testing first, at small scale, catches that risk early and cheaply, extending G7's Hypothesis Testing to bigger stakes.
- The assumptions that feel most obviously true are exactly the ones most worth testing, not the ones to skip.

COMMON MISCONCEPTIONS TO PREEMPT

- Skipping the small test because you're confident the assumption is obviously true.
- Treating high confidence in an assumption as a reason not to verify it.

SUGGESTED TIME: 20 min

FIRST H30

Strategic Mapping

LEARNING OBJECTIVE

Participants build an explicit map connecting their current near-term actions to a genuine longer-term goal.

KEY TALKING POINTS

- Without an explicit map connecting near-term actions to longer-term goals, daily priorities can drift away from what actually matters.
- This closes the entire stage by combining prioritization, systems thinking, and scenario planning into one integrated view.
- A map is only useful if revisited periodically — flag this explicitly as an ongoing practice, not a one-time artifact.

COMMON MISCONCEPTIONS TO PREEMPT

- Building the map once and never revisiting it as circumstances change.
- Letting the map become outdated and continuing to reference it anyway.

SUGGESTED TIME: *25 min*